Vignette for CopulaSCR package

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Abstract

We propose a novel copula-based framework that learns dependence among multiple correlated marginal components through a pseudo-likelihood for estimation. We adopt nonparametric marginals, alleviating the reliance on marginal distribution assumptions typically required in conventional copula models, and estimate the association between the onsets of intermediate cardiovascular diseases and death by solving a concordance estimating equation. Under this framework, a renewable risk assessment method is developed for dynamic survival prediction, leveraging information on disease onset times and the maximum follow-up duration. Methodological details can be found in the papar on "Learning association from multiple intermediate events for dynamic prediction of survival: an application to cardiovascular disease prognosis" by Tonghui Yu and Liming Xiang.

Keywords: Copula, Informative censoring, multiple intermediate events, terminal prediction.

Generate data with $K(=3)$ intermediate events and a terminal event under Frank Copula structure:

```
R> library(CopulaSCR)
R> set.seed(12345)
R> mdata<- simSCRmul(n=100,copulafam = "frank",K=3, tau.alpha = .5,tau.theta=.5)
```

Learn association from multiple intermediate events and terminal event under Frank Copula structure:

```
R> mfit<- mscr(mT1=mdata$T1,T2=mdata$T2,mevent1=mdata$event1,event2=mdata$event2,
+     copulafam = "frank",ncore = 1)
R> print(mfit)

Call: mscr(mT1 = mdata$T1, T2 = mdata$T2, mevent1 = mdata$event1,
    event2 = mdata$event2, copulafam = "frank", ncore = 1)

3 intermediate events, 1 terminal event.
-------
Association analyses
Estimated Kendall's tau in the FRANK copula structure
tau.theta:
   T1    T2    T3
0.574 0.592 0.563
```

```
tau.alpha:
[1] 0.456
-------
Marginal analyses (survival curves)
intermediate event times: A list of 3 scrsurv objects.
terminal event time:
Step function with continuity 'f'= 0 ,  100 knots with summary
     Min.   1st Qu.    Median      Mean   3rd Qu.      Max.
 0.007815  0.485548  0.896666  1.570410  1.918267 10.670315

and        101 plateau levels (y) with summary
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 0.0000  0.3061  0.5321  0.5320  0.7769  1.0000
```

Learn association from multiple intermediate events and terminal event under Clayton Copula structure, and Compare with the results under Frank Copula structure:

```
R> mfit_c<- mscr(mT1=mdata$T1,T2=mdata$T2,mevent1=mdata$event1,event2=mdata$event2,
+     copulafam = "clayton",ncore = 1)
R> mfit$logLik.alpha>mfit_c$logLik.alpha

[1] TRUE
```

This suggests the frank copula structure is the best and indeed the truth.

Plot the association structure within the data:

```
R> plot(mfit,type = "data")
```

Plot the association structure within the data:

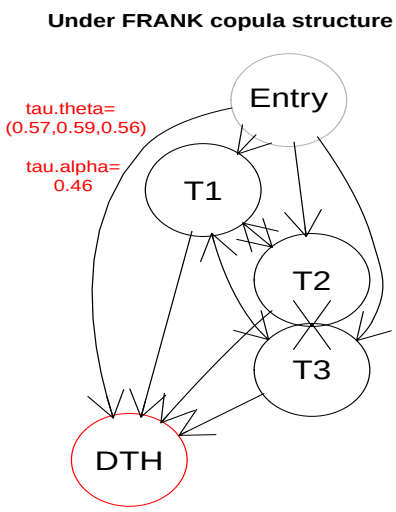

Figure 1: Data structure of the `mdata` sample.

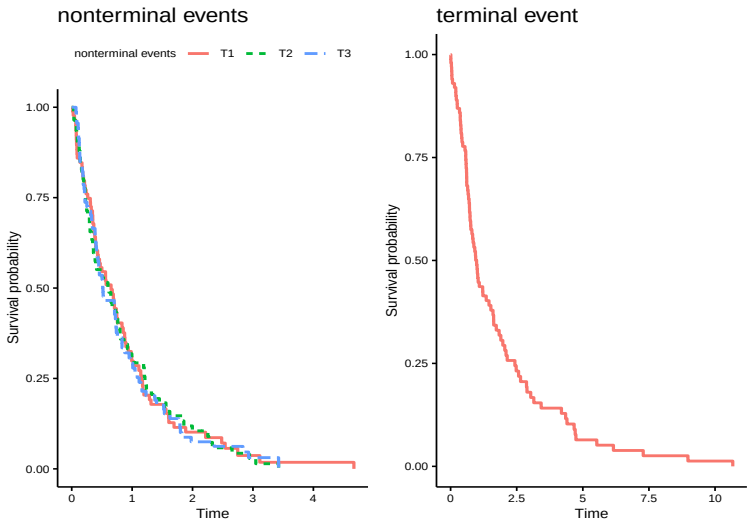

Figure 2: Marginal survival curves of the `mdata` sample.

```
R> plot(mfit,type = "msurv", nt.seperate = FALSE)
```

Survival prediction:

```
R> mT1<- mdata$T1
R> mT1[mdata$event1==0]<- NA
R> mT1.te<- mT1[1:4,]
R> ####  dynamic prediction for conditional restricted mean survival time
R> rmst<- predict(mfit, t1obs=mT1.te,type="rmst")
R> rmst<- sapply(rmst$rmst, function(x)x$rmst)
R> ## naive E(D|D> t_m)
R> rmst0<- predict(mfit, t1obs=mT1.te,type="rmst",cause=0)
R> rmst0<- sapply(rmst0$rmst, function(x)x$rmst)
R> ## E(D|T_1=t_1,D> t_1)
R> rmst1<- predict(mfit, t1obs=mT1.te,type="rmst",cause=1,maxobs = FALSE)
R> rmst1<- sapply(rmst1$rmst, function(x)x$rmst)
R> ## E(D|T_1=t_1,D> t_m)
R> rmst1m<- predict(mfit, t1obs=mT1.te,type="rmst",cause=1,maxobs = TRUE)
R> rmst1m<- sapply(rmst1m$rmst, function(x)x$rmst)
```

To compute conditional quantile survival time (CQST), use argument `type = "qst"` with some specified quantile level, for example, `tau = 0.1` corresponding to 0.1 quantile.

Accuracy measure (MSE):

```
R> T2<- mdata$T2
R> T2[mdata$event2==0]<- NA
R> T2<- T2[1:4]
R> mse<- apply(cbind(T2-rmst,T2-rmst0,T2-rmst1,rmst1m)^2,2,mean)
```

```
R> names(mse)<- c("DP","P0","P1","P1m")
R> print(round(mse,2))

   DP    P0    P1   P1m
 0.48  2.36 70.37 85.39
```

Brier Score:

```
R> mT1.te<- mdata$T1
R> mT1.te[mdata$event1==0]<- NA
R> ####  dynamic prediction
R> surv<- predict(mfit, t1obs=mT1.te,type="surv")
R> ## naive Pr(D>t|D> t_m)
R> surv0<- predict(mfit, t1obs=mT1.te,type="surv",cause=0)
R> ## Pr(D>t|T_1=t_1,D> t_1)
R> surv1<- predict(mfit, t1obs=mT1.te,type="surv",cause=1,maxobs = FALSE)
R> ## Pr(D>t|T_1=t_1,D> t_m)
R> surv1m<- predict(mfit, t1obs=mT1.te,type="surv",cause=1,maxobs = TRUE)
R> ### Brier Scores
R> BS<- dyBS(surv, S_D = mfit$S_D,T2 = mdata$T2,
+     event2 = mdata$event2, int.method = "natural")
R> BS0<- dyBS(surv0, S_D = mfit$S_D,T2 = mdata$T2,
+     event2 = mdata$event2, int.method = "natural")
R> BS1<- dyBS(surv1, S_D = mfit$S_D,T2 = mdata$T2,
+     event2 = mdata$event2, int.method = "natural")
R> BS1m<- dyBS(surv1m, S_D = mfit$S_D, T2 = mdata$T2,
+     event2 = mdata$event2, int.method = "natural")
R> ibs<-c(BS$IBS,BS0$IBS,BS1$IBS,BS1m$IBS) ## integrated BS
R> names(ibs)<- c("DP","P0","P1","P1m")
R> print(round(ibs,2))

  DP   P0   P1  P1m
0.59 1.02 0.59 0.59

R> plot(BS$BS,ylim = c(0,0.7),type="l",xlab="time",ylab="Brier Score",col=1)
R> lines(BS0$BS,col=2)
R> lines(BS1$BS,col=3)
R> lines(BS1m$BS,col=4)
R> legend("topright",col = 1:4,lty = 1,legend = c("DP","P0","P1","P1m"))
```

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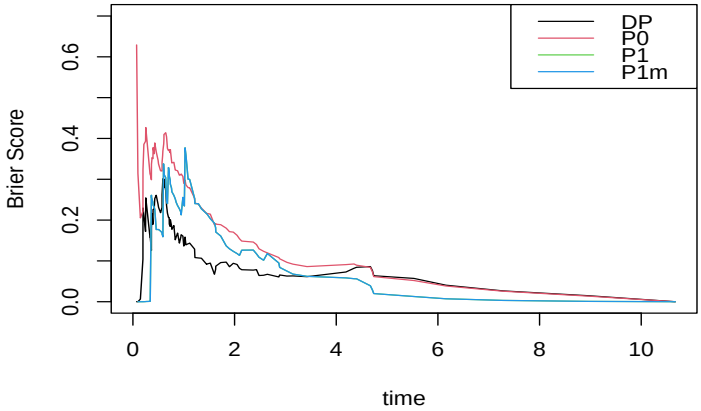

Figure 3: Brier Scores for the `mdata` sample using different prediction algorithm.

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